

7

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks 20

Annexure A

Descriptive Written Test

Code of Conduct:

I HEMASRINITHA.P. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Hemasrinitha P.

16
20

B

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CO1 - AT3 - DESCRIPTIVE

WRITING TEST.

1. A smart home automation system automatically controls lighting, temperature and security using sensors such as motion detectors, temperature sensors and timers. It learns about user habits to improve energy efficiency and security.

Some of the different types of intelligent agents are as follows:

SIMPLE REFLEX AGENT -

The agent that responds only to the current input using predefined rules are called simple reflex agent. It is fast but cannot learn from past events. Examples: Lights are turned on, when motion is detected by the motion sensors.

MODEL-BASED AGENT -

The agent that stores previous information and uses it to make better decisions is called Model-Based agent.

Examples: Remember user's routine and adjusts lighting or temperature accordingly.

GOAL-BASED AGENT:

A goal-based agent makes decisions to achieve a specific goals such as maintaining a comfortable temperature or reducing electricity consumption.

UTILITY-BASED AGENT:

This agent chooses the action that provides the maximum overall benefit by balancing comfort, energy efficiency, cost and safety.

Example: It cools only occupied rooms to save electricity consumption.

CONCLUSION:

A hybrid approach that combines Model-Based goal-based and utility-based agents is the most effective because it remembers past preferences, plan actions, and makes optimized decisions, that ensures better comfort, energy efficiency and security.

2. A robot placed in an unknown maze must find a path to exit without knowing the layout in advance. Uninformed Linear search algorithms help the robot explore the maze and reach the goal.

UNIFORM COST SEARCH (UCS):

It expands the path with lowest cumulative cost first. It guarantees the least cost path when movement costs are different.

LITERATURE DEEPENING SEARCH (IDS):

It repeatedly performs depth-limited search with increasing depth until the exit is found. It uses very little memory. It revisits the nodes, increasing search time.

IDS is optimal and suitable for mazes with different path costs, while IDS is memory-efficient and works well when all moves have equal cost.

GOAL-BASED AGENTS:

It is most suitable to reach the exit by its action. It revisits the same locations.

CONCLUSION:

A combination of Goal-Based Agent and uniform cost search (UCS) is the suitable choice because it helps the robot intelligently plan its actions and leads optimal path to the exit. If memory is limited, Goal-Based Agent with IDS is a good alternative choice.